

TIC 232087808

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS

Period 3.726465 d
Epoch (t0) 1468.4912 BTJD
Duration 0.1200 d
Depth 4.873 % (48,726 ppm)
SNR 108.8

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

Eclipses observed 5 (gate ≥ 3)
Robust depth ratio 1.04 (gate ≥ 0.50)
Robust depth sigma 89.9 (gate ≥ 7.0)
Odd/even z-score 0.91 (consistent)
Secondary eclipse detected, phase -0.49, 5.3 sigma

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 6 light-curve dip in this sector: 91.7 sigma t0 refined by -0.0050 phase
Difference-image peak SNR 44.10 (gate ≥ 5.0) Compactness 1.00 (gate ≥ 0.50)
Dimming centroid offset 0.17 px (4 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 232087810
0.53 px away, dTmag +2.26 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)

